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Semiconductor device

Abstract

In a semiconductor device, a semiconductor substrate has an element region and a peripheral region, and trenches are defined on an upper surface of the semiconductor substrate. The trenches extend in a first direction, and are arranged at intervals in a second direction. The element region includes an n-type source region, a p-type contact region, a p-type body region, an n-type drift region, a p-type bottom region, and p-type connection regions. The bottom region is spaced from a bottom surface of the trenches. The connection regions connect the body region and the bottom region, extend in the first direction, and are arranged at intervals in the second direction. The element region has outer side portions and a central portion in the second direction. An interval between the connection regions in the second direction is greater in the outer side portion than in the central portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) The present application claims the benefit of priority from Japanese Patent Application No. 2022-035672 filed on Mar. 8, 2022. The entire disclosures of the above application are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a semiconductor device.

BACKGROUND

(3) For example, there is a semiconductor device having a semiconductor substrate, a gate insulating film, and a gate electrode. The semiconductor substrate has multiple trenches on its upper surface, and the gate insulating film is disposed to cover an inner surface of each trench. The gate electrode is disposed on the gate insulating film in each of the trenches. The semiconductor substrate includes an n-type source region, a p-type contact region, a p-type body region, an n-type drift region, a p-type bottom region, and multiple p-type connection regions. The source region is exposed on the upper surface of the semiconductor substrate and is in contact with the gate insulating film. The contact region is exposed on the upper surface of the semiconductor substrate. The body region is in contact with the gate insulating film below the source region and is in contact with the contact region. The drift region is in contact with the gate insulating film below the body region. The bottom region is located below the trench and is spaced from the bottom surface of the

trench. Each of the connection regions connects the body region and the bottom region. The connection regions extend parallel to the trenches, and are spaced apart from each other in a direction perpendicular to the direction in which the trenches extend.

SUMMARY

(4) The present disclosure describes a semiconductor device which is capable of reducing switching loss while reducing an on-resistance. According to an aspect of the present disclosure, a semiconductor device includes a semiconductor substrate that has an element region and a peripheral region on a periphery of the element region, a plurality of trenches defined on an upper surface of the semiconductor substrate, a gate insulating film covering an inner surface of each trench, and a gate electrode disposed in the trench and separated from the semiconductor substrate by the gate insulating film. The trenches extend in a first direction along the upper surface of the semiconductor substrate, and are arranged at intervals in a second direction perpendicular to the first direction along the upper surface of the semiconductor substrate. The element region includes an n-type source region, a p-type contact region, a p-type body region, an n-type drift region, a p-type bottom region, and p-type connection regions. The bottom region is spaced from a bottom surface of the trench. The connection regions connect the body region and the bottom region, extend in the first direction, and are arranged at intervals in the second direction. The element region has outer side portions and a central portion defined between the outer side portions in the second direction. An interval between the connection regions in the second direction is greater in at least one of the outer side portions than in the central portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Objects, features and advantages of the present disclosure will become more apparent from the following detailed description made with reference to the accompanying drawings, in which like parts are designated by like reference numbers and in which:
- (2) FIG. 1 is a plan view of a semiconductor device according to a first embodiment of the present disclosure;
- (3) FIG. 2 is an enlarged plan view of a part of the semiconductor device including a boundary between an element region and a peripheral region;
- (4) FIG. 3 is a cross-sectional view taken along a line III-III of FIG. 2;
- (5) FIG. 4 is a cross-sectional view taken along a line IV-IV of FIG. 2;
- (6) FIG. 5 is an enlarged plan view of a part of a semiconductor device including a boundary between an element region and a peripheral region according to a second embodiment of the present disclosure;
- (7) FIG. 6 is an enlarged plan view of a part of a semiconductor device including a boundary between an element region and a peripheral region according to a third embodiment of the present disclosure;
- (8) FIG. 7 is a cross-sectional view taken along a line VII-VII of FIG. 6;
- (9) FIG. 8 is a cross-sectional view taken along a line VIII-VIII of FIG. 6;
- (10) FIG. 9 is an enlarged plan view of a part of a semiconductor device including a boundary between an element region and a peripheral region according to a fourth embodiment of the present disclosure;
- (11) FIG. 10 is a cross-sectional view taken along a line X-X of FIG. 9;
- (12) FIG. 11 is a cross-sectional view, corresponding to FIG. 3, of a semiconductor device according to a modification; and
- (13) FIG. 12 is a cross-sectional view, corresponding to FIG. 3, of a semiconductor device according to another modification.

DETAILED DESCRIPTION

(14) To begin with, a relevant technology will be described only for understanding the embodiments of the present disclosure.

(15) For example, there is a semiconductor device having trenches in a semiconductor substrate and in which the semiconductor substrate includes an n-type source region, a p-type contact region, a p-type body region, an n-type drift region, a p-type bottom region, and multiple p-type connection regions. The source region is exposed on an upper surface of the semiconductor substrate and is in contact with the gate insulating film. The contact region is exposed on the upper surface of the semiconductor substrate. The body region is in contact with a gate insulating film disposed in the trench below the source region and is in contact with the contact region. The drift region is in contact with the gate insulating film below the body region. The bottom region is located below the trench and is spaced from the bottom surface of the trench. Each of the connection regions connects the body region and the bottom region. The connection regions extend parallel to the trenches, and are spaced apart from each other in a direction perpendicular to the direction in which the trenches extend.

(16) When such a semiconductor device is turned off, a depletion layer extends from the bottom region into the drift region. The depletion layer extending from the bottom region into the drift region suppresses electric field concentration at a bottom end of the trench.

(17) In such a semiconductor device, a p-n diode (hereinafter referred to as a body diode) is parasitically formed by the p-type contact region and body region and the n-type drift region. In the operation of such a semiconductor device, when a forward biased voltage is applied to the body diode, the body diode is turned on and holes flow from the contact region into the drift region via the body region. Thereafter, when the voltage applied to the body diode is switched to a reverse biased voltage, the holes accumulated in the drift region flow into the contact region via the body region in the process of turning off the body diode. That is, a recovery current occurs.

(18) In such a semiconductor device, the bottom region is provided inside the drift region, and is connected to the body region via the connection regions. Therefore, during reverse biasing, the holes accumulated in the drift region readily flow from the drift region to the bottom region, and thus most of the holes flow to the body region via the bottom region and the connection regions.

(19) When such a semiconductor device is turned on, the depletion layer that has extended from the connection region to the drift region contracts. In a period before the depletion layer completely contracts, the depletion layer restricts the path through which the main current of the semiconductor device flows. If the connection regions are densely arranged, the on-resistance is high immediately after the semiconductor device is turned on.

(20) If the connection regions are arranged not densely in order to reduce the on-resistance, the current density in each connection region is likely to be high when the holes accumulated in the drift region flows toward the body region at a time the voltage applied to the body diode is switched to the reverse biased voltage. Therefore, the electrical resistance increases when the recovery current flows, and the temperature of the semiconductor device is likely to rise. As a result, switching loss increases.

(21) The present disclosure provides a semiconductor device which is capable of reducing the switching loss while reducing the on-resistance.

(22) In an embodiment of the present disclosure, a semiconductor device includes: a semiconductor substrate, a plurality of trenches, a gate insulating film, and a gate electrode. The semiconductor substrate has an element region and a peripheral region disposed on a periphery of the element region. The plurality of trenches is defined on an upper surface of the semiconductor substrate. Each of the plurality of trenches extends in a first direction along the upper surface of the semiconductor substrate. The plurality of trenches is arranged spaced apart from each other in a second direction that is along the upper surface of the semiconductor substrate and perpendicular to the first direction. The gate insulating film covers an inner surface of each of the plurality of

trenches. The gate electrode is disposed in each of the plurality of trenches and insulated from the semiconductor substrate by the gate insulating film. The element region includes, an n-type source region, a p-type contact region, a p-type body region, an n-type drift region, a p-type bottom region, and a plurality of p-type connection regions. The n-type source region is exposed on the upper surface of the semiconductor substrate and is in contact with the gate insulating film disposed in a corresponding trench. The p-type contact region is exposed on the upper surface of the semiconductor substrate. The p-type body region is in contact with the gate insulating film disposed in the corresponding trench at a position below the source region, and is in contact with the contact region. The n-type drift region is in contact with the gate insulating film disposed in the corresponding trench at a position below the body region, and is separated from the source region by the body region. The p-type bottom region is disposed below the corresponding trench and spaced from a bottom surface of the corresponding trench, and is surrounded by the drift region. Each of the plurality of p-type connection regions connects the body region and the bottom region, and extends in the first direction. The plurality of connection regions is arranged spaced apart from each other in the second direction. The element region has outer side portions at opposite ends of the element region in the second direction, and a central portion defined between the outer side portions. An interval in the second direction between the connection regions provided in at least one of the outer side portions is greater than an interval in the second direction between the connection regions provided in the central portion.

(23) In such a semiconductor device, the element region has the outer side portions at opposite ends of the element region in the second direction in which the trenches and the connection regions are arranged, and a central portion between the outer side portions. The interval between the connection regions in at least one of the outer side portions is larger than the interval between the connection regions in the central portion. In the at least one of the outer side portions, since the connection regions are widely spaced apart from each other, the on-resistance is low. Therefore, the on-resistance of the element region as a whole can be reduced. In addition, the outer side portion is adjacent to the peripheral region. In the peripheral region, since a voltage is less likely to be applied to a body diode, holes are hardly accumulated in the drift region during forward biasing. That is, since few holes flow into the outer side portion from the peripheral region, the density of holes accumulated in the drift region during the forward biasing is low in the outer side portion. Therefore, the recovery current is small in the outer side portion, and the temperature of the semiconductor device hardly rises even if the interval between the connection regions in the outer side portion is large. For this reason, the switching loss is small in the element region as a whole. Accordingly, in the semiconductor device, switching loss can be reduced while reducing the on-resistance.

(24) In an embodiment of the present disclosure, in the semiconductor device, the interval in the second direction between the connection regions provided in the second direction in the outer side portion may be increased toward the end of the element region.

(25) The amount of holes accumulated in the drift region when the body diode is forward biased decreases toward the end of the element region. In the configuration described above, since the interval between the connection regions is increased toward the end of the element region, it is possible to more preferably achieve the reduction in on-resistance and the reduction in switching loss.

(26) In an embodiment of the present disclosure, in the semiconductor device, each of the plurality of bottom regions may extend in the second direction, and the plurality of bottom regions may be spaced apart from each other in the first direction.

(27) In such a configuration, the direction in which the bottom region extends and the direction in which the connection region extends intersect to each other. Thus, the bottom region and the connection region can be connected more reliably than, for example, a configuration in which the bottom region and the connection region extend parallel to each other.

(28) In an embodiment of the present disclosure, in the semiconductor device, each of the outer side portions may further include a plurality of p-type auxiliary connection regions. The plurality of auxiliary connection regions may each connect the body region and the bottom region, and be arranged spaced apart from each other in the first direction.

(29) In such a configuration, in the outer side portion, holes flow from the bottom region to the body region via the auxiliary connection region in addition to the connection region when the body diode is reverse biased. In this case, since the holes can flow while branching in many paths, switching loss can be further reduced. In addition, since the auxiliary connection regions are spaced apart in the first direction, the range of the depletion layer extending from the auxiliary connection region to the drift region is narrow, and the path of the main current is less likely restricted when the semiconductor device is turned on. Therefore, the increase in on-resistance can be suppressed.

(30) In an embodiment of the present disclosure, in the semiconductor device, an interval defined between the trenches in the second direction in the outer side portion may be smaller than an interval defined between the trenches in the second direction in the central portion.

(31) In such a configuration, since the channel density is increased in the outer side portions, the channel resistance, that is, the on-resistance can be reduced.

(32) Embodiments of the present disclosure will be described further in detail with reference to the drawings.

First Embodiment

(33) A semiconductor device **10** of a first embodiment is shown in FIGS. **1** to **4**. The semiconductor device **10** is a metal oxide semiconductor field effect transistor (MOSFET). As shown in FIG. **1**, the semiconductor device **10** has a semiconductor substrate **12**. The semiconductor substrate **12** has an element region **62** and a peripheral region **64**. Components of the MOSFET are formed inside the semiconductor substrate **12** in the element region **62**. The peripheral region **64** is arranged around the element region **62**. Although not shown, a peripheral voltage withstand structure, such as a guard ring, is formed inside the semiconductor substrate **12** in the peripheral region **64**. For example, the semiconductor substrate **12** is made of silicon carbide (SiC). However, the material of the semiconductor substrate **12** is not particularly limited, and may be other semiconductor materials such as silicon (Si) or gallium nitride (GaN). Hereinafter, one direction parallel to an upper surface **12a** of the semiconductor substrate **12** is referred to as an x direction, and a direction parallel to the upper surface **12a** of the semiconductor substrate **12** and perpendicular to the x direction is referred to as a y direction. Also, a thickness direction of the semiconductor substrate **12** is referred to as the z direction. The z direction is perpendicular to the x direction and the y direction.

(34) As shown in FIG. **1**, in the element region **62**, trenches **22** are defined in the upper surface **12a** of the semiconductor substrate **12**. Each of the trenches **22** extends long along the y direction. Thus, the y direction corresponds to a longitudinal direction of the trench **22**. The multiple trenches **22** extend parallel to each other. The trenches **22** are arranged at intervals in the x direction. FIG. **2** is an enlarged view of a part shown in FIG. **1** including a boundary between the element region **62** and the peripheral region **64** adjoined in the x direction. Note that, in FIGS. **1** and **2**, illustration of the components above the upper surface **12a** of the semiconductor substrate **12**, such as an insulating film and an electrode, is omitted. As shown in FIGS. **2** to **4**, a gate insulating film **24** and a gate electrode **26** are provided in each trench **22**. The gate insulating film **24** covers the inner surface of each trench **22**. The gate electrode **26** is disposed inside each trench **22**. The gate electrode **26** is insulated from the semiconductor substrate **12** by the gate insulating film **24**.

(35) As shown in FIGS. **3** and **4**, the top surface of each gate electrode **26** is covered with an interlayer insulating film **28**. In the peripheral region **64**, an insulating film **65** covers substantially the entire upper surface **12a** of the semiconductor substrate **12**. An upper electrode **70** is arranged on the upper surface **12a** of the semiconductor substrate **12**. In the element region **62**, the upper electrode **70** is in contact with the upper surface **12a** of the semiconductor substrate **12** at locations

where the interlayer insulating film **28** is not provided. The upper electrode **70** is insulated from the gate electrode **26** by the interlayer insulating film **28**. A lower electrode **72** is arranged on a lower surface **12b** of the semiconductor substrate **12**. The lower electrode **72** is in contact with substantially the entire lower surface **12b** of the semiconductor substrate **12**.

(36) The element region **62** is provided with multiple source regions **30**, multiple contact regions **31**, a body region **32**, a drift region **34**, a drain region **35**, multiple bottom regions **36**, and multiple connection regions **38**.

(37) Each of the source regions **30** is an n-type region. The source region **30** is provided at a position exposed on the upper surface **12a** of the semiconductor substrate **12**. The source region **30** is in ohmic contact with the upper electrode **70**. The source region **30** is disposed at the side surface of the trench **22**, and is in contact with the gate insulating film **24** disposed inside the trench **22**.

(38) Each of the contact regions **31** is a p-type region. The contact region **31** is provided at a position exposed on the upper surface **12a** of the semiconductor substrate **12**. The contact region **31** is arranged at a position interposed between two source regions **30**. The contact region **31** is in ohmic contact with the upper electrode **70**.

(39) The body region **32** is a p-type region. The body region **32** has a p-type impurity concentration lower than that of the contact region **31**. The body region **32** is in contact with the bottom of the source region **30** and the bottom of the contact region **31**. The body region **32** is in contact with the gate insulating film **24** disposed in each of the trenches **22** below the source regions **30**. The body region **32** is arranged across the element region **62** and the peripheral region **64**.

(40) The drift region **34** is an n-type region. The drift region **34** is arranged below the body region **32**. The drift region **34** is in contact with the bottom of the body region **32**. The drift region **34** is in contact with the gate insulating film **24** disposed in each of the trenches **22** below the body region **32**. The drift region **34** is separated from the source regions **30** by the body region **32**. The drift region **34** is arranged across the element region **62** and the peripheral region **64**.

(41) The drain region **35** is an n-type region. The drain region **35** is arranged below the drift region **34**. The drain region **35** has an n-type impurity concentration higher than that of the drift region **34**. The drain region **35** is in contact with the bottom of the drift region **34**. The drain region **35** is exposed on the lower surface **12b** of the semiconductor substrate **12**. The drain region **35** is in ohmic contact with a lower electrode **72** disposed on the lower surface **12b** of the semiconductor substrate **12**. The drain region **35** is arranged across the element region **62** and the peripheral region **64**.

(42) Each of the bottom regions **36** is a p-type region. Each of the bottom regions **36** extends in a direction perpendicular to the longitudinal direction of the trenches **22**, that is, extends in the x direction. As shown in FIG. 2, the bottom regions **36** are arranged at intervals in the y direction. Each of the bottom regions **36** is positioned below the trenches **22** and spaced from the bottom surfaces of the trenches **22**. A periphery of each bottom region **36** is surrounded by the drift region **34**. Each of the bottom regions **36** extends from the element region **62** into the peripheral region **64**, that is, extends across the element region **62** and the peripheral region **64**.

(43) Each of the connection regions **38** is a p-type region. As shown in FIG. 2, each of the connection regions **38** extends parallel to the trenches **22**, that is, extends in the y direction. The connection regions **38** are arranged spaced apart from each other in the direction perpendicular to the longitudinal direction of the trenches **22**. Each of the connection regions **38** is arranged in a region between adjacent two trenches **22** when viewed from the top, that is, in a plan view shown in FIG. 2. As shown in FIG. 3, each of the connection regions **38** connects the body region **32** and the bottom region **36**. The connection regions **38** are provided also in the peripheral region **64**.

(44) As noted above, each of the bottom regions **36** is connected to the body region **32** via the respective connection regions **38**. Therefore, each of the bottom regions **36** is connected to the upper electrode **70** through the connection regions **38**, the body region **32** and the contact region **31**. As such, the potential of each bottom region **36** is approximately equal to the potential of the

upper electrode **70**.

(45) As shown in FIG. 1, the element region **62** has two outer side portions **62a** and a central portion **62b**. The outer side portions **62a** are located at opposite ends of the element region **62** in the x direction. Although only one end of the element region **62** in the x direction is shown in FIGS. 2 to 4, the other outer side portion **62a** is located at the other end of the element region **62** in the x direction. The central portion **62b** is located between the two outer side portions **62a**.

(46) As shown in FIG. 3, each connection region **38** is provided in a respective range defined between the adjacent trenches **22** in the central portion **62b**. That is, in the central portion **62b**, the connection regions **38** and the trenches **22** are alternately arranged along the x direction. On the other hand, the connection region **38** is not provided in a range defined between the adjacent two trenches **22** in the outer side portion **62a**. The connection regions **38** are provided in the peripheral region **64** at the same intervals as those in the central portion **62b**. Therefore, an interval **d1** defined between the connection regions **38** in the outer side portion **62a** in the x direction is greater than an interval **d2** between the connection regions **38** in the central portion **62b** in the x direction.

Although not shown, the interval between the connection regions **38** is similarly set in the other outer side portion **62a** as well. The interval between two adjacent trenches **22** is approximately equal between the outer side portions **62a** and the central portion **62b**.

(47) When the semiconductor device **10** is used, the lower electrode **72** is applied with a higher potential than that to the upper electrode **70**. When a voltage equal to or higher than a gate threshold is applied to the gate electrode **26**, a channel is formed in the body region **32** in a range in contact with the gate insulating film **24**, and the semiconductor device **10** is turned on. When the voltage applied to the gate electrode **26** is lowered below the gate threshold, the channel disappears and the semiconductor device **10** is turned off.

(48) When the semiconductor device **10** is in an off state, the potential of the lower electrode **72** is much higher than the potential of the upper electrode **70**. In this state, the drift region **34** has a potential approximate to the potential of the lower electrode **72**. Also, as described above, the bottom region **36** has a potential approximately the same as the potential of the upper electrode **70**. Therefore, a high reverse voltage is applied to the p-n junction at the interface between the drift region **34** and the bottom region **36**. With this, a depletion layer extends widely from each bottom region **36** into the drift region **34**. As a result, electric field concentration at positions near the lower ends of the trenches **22** is suppressed, and the breakdown voltage of the semiconductor device **10** is ensured. The reverse voltage is also applied to the p-n junction at the interface between the connection region **38** and the drift region **34**. Therefore, the depletion layer extends from the connection region **38** into the drift region **34** as well.

(49) When the semiconductor device **10** is turned on, holes are supplied from the upper electrode **70** to each bottom region **36** via the contact regions **31**, the body region **32** and the connection regions **38**. As a result, the depletion layer extending from the connection region **38** and the bottom region **36** into the drift region **34** contracts. In a period before the holes are supplied to the connection region **38** and the bottom region **36**, the depletion layer is extended from the connection region **38** and the bottom region **36** into the drift region **34**. Therefore, the path of a main current is restricted by the depletion layer immediately after the semiconductor device **10** is turned on. In the present embodiment, however, the interval **d1** between the connection regions **38** at the outer side portion **62a** of the element region **62** is widened. Therefore, a non-depletion portion relatively widely exists in the drift region **34** in the outer side portion **62a**. As such, immediately after the semiconductor device **10** is turned on, the relatively wide range of the drift region **34**, in particular, the drift region **34** located between the trenches **22** in the outer side portion **62a** can be used for the path of the main current. Therefore, in the semiconductor device **10** of the present embodiment, the on-resistance of the element region **62** as a whole can be reduced.

(50) Inside the semiconductor substrate **12**, a p-n diode (hereinafter referred to as a body diode) is parasitically formed by the p-type contact region **31** and body region **32** and the n-type drift region

34 and drain region **35**. In the operation of the semiconductor device **10**, there is a case where the upper electrode **70** is applied with the higher potential than that of the lower electrode **72**. When the potential of the upper electrode **70** is higher than that of the lower electrode **72**, that is, when in forward bias, the body diode is turned on. That is, holes flow from the upper electrode **70** into the drift region **34** via the contact region **31** and the body region **32**.

(51) Thereafter, when the potential of the lower electrode **72** is switched to a higher potential than the upper electrode **70**, that is, when in reverse bias, the holes accumulated in the drift region **34** flows to the upper electrode **70** via the body region **32** and the contact region **31** in a process where the body diode is turned off. That is, a recovery current flows. The semiconductor device **10** has the multiple bottom regions **36** inside the drift region **34**, and the bottom regions **36** are connected to the body region **32** through the connection regions **38**. Therefore, during the reverse biasing, the holes accumulated in the drift region **34** easily flow from the drift region **34** to the bottom regions **36**, and thus most of the holes flow to the body region **32** via the bottom regions **36** and the connection regions **38**.

(52) In the present embodiment, the interval **d2** between the connection regions **38** in the central portion **62b** of the element region **62** is narrow. The interval **d2** is smaller than the interval **d1**. That is, the connection regions **38** are densely arranged in the central portion **62b**. Therefore, there are many paths for holes to flow from the bottom regions **36** to the body region **32**, and thus the holes are quickly discharged to the upper electrode **70**. As a result, the switching loss generated in the central portion **62b** is small, and the temperature rise of the central portion **62b** is suppressed. On the other hand, the interval **d1** between the connection regions **38** in the outer side portion **62a** of the element region **62** is wide. Therefore, there are fewer paths for holes to flow from the bottom regions **36** to the body region **32**, as compared to the central portion **62b**. However, the outer side portion **62a** is adjacent to the peripheral region **64**. Since the contact region **31** does not exist in the peripheral region **64**, holes are less likely to be accumulated in the drift region **34** of the peripheral region **64** during the forward biasing. Therefore, the density of holes accumulated in the drift region **34** during the forward biasing is lower in the outer side portion **62a** than in the central portion **62b**. With this, the recovery current in the outer side portion **62a** is small. As such, even if the interval **d1** between the connection regions **38** in the outer side portion **62a** is wide, the density of the recovery current flowing through each connection region **38** is not so high. As a result, the switching loss generated in the outer side portion **62a** is small, and the temperature rise of the outer side portion **62a** is suppressed. In the present embodiment, as described above, when the recovery current flows, the temperature rise is suppressed in the element region **62** as a whole, and the switching loss is suppressed.

(53) As described above, when the semiconductor device **10** is turned off, the depletion layer extends from the body region **32**, the connection regions **38** and the bottom regions **36** to the drift region **34**. As a result, substantially the entire drift region **34** is depleted. When a high voltage is applied to the lower electrode **72** while the semiconductor element **10** is in the off state, an avalanche current flows from the drift region **34** to the upper electrode **70** via the bottom regions **36**, the connection regions **38**, the body region **32** and the contact regions **31**. In the central portion **62b**, since the interval between the connection regions **38** is small, the density of the avalanche current flowing through each connection region **38** is low. As such, the temperature rise in the central portion **62b** is suppressed. Further, in the outer side portion **62a**, since the interval between the connection regions **38** is large, the density of the avalanche current flowing through each connection region **38** is high. However, since the outer side portion **62a** is adjacent to the peripheral region **64**, the outer side portion **62a** has a higher heat dissipation property. Therefore, the temperature rise of the outer side portion **62a** is suppressed. In this way, when the avalanche current occurs, the temperature rise is suppressed in each of the central portion **62b** and the outer side portion **62a**. Thus, in the present embodiment, the temperature rise is suppressed in the element region **62** as a whole when the avalanche current occurs. As such, the semiconductor

device **10** has high avalanche resistance.

(54) Although the semiconductor substrate **12** generates heat during operation of the semiconductor device **10**, the heat is more easily released in the outer side portion **62a** of the element region **62** than in the central portion **62b**. The outer side portion **62a** is adjacent to the peripheral region **64** in which the main current does not flow. Therefore, the temperature of the outer side portion **62a** is less likely to rise than that of the central portion **62b**, and the temperature of the outer side portion **62a** is lower than the temperature of the central portion **62b**. As such, even if the temperature rises in the outer side portion **62a**, its influence is small. In the semiconductor device **10** of the present embodiment, as described above, an increase in switching loss and a decrease in avalanche resistance can be suppressed while reducing the on-resistance.

(55) In the present embodiment, the direction (x direction) in which the bottom region **36** extends and the direction (y direction) in which the connection region **38** extends intersect to each other. Therefore, as compared to the configuration in which the bottom region **36** and the connection region **38** extend parallel to each other, the bottom region **36** and the connection region **38** can be connected more reliably.

Second Embodiment

(56) A semiconductor device **100** of a second embodiment differs from the semiconductor device **10** of the first embodiment in the configuration of the connection region **38**. In the second embodiment, the interval between the connection regions **38** in the outer side portion **62a** increases toward the end of the element region **62**. As shown in FIG. 5, in the second embodiment, the outer side portion **62a** has nine trenches **22**. Further, the outer side portion **62a** has four connection regions **38**. In the second embodiment, the width of the outer side portion **62a** in the x direction is greater than that in the first embodiment. In the following description, for the sake of convenience, the connection regions **38** provided in the outer side portion **62a** are labelled as first connection region **38a**, second connection region **38b**, third connection region **38c** and fourth connection region **38d** in order from the outermost one that is the closest to the outer peripheral region **64** toward the central portion **62b**. As shown in FIG. 5, an interval **d3** between the first connection region **38a** and the second connection region **38b** is larger than an interval **d4** between the second connection region **38b** and the third connection region **38c**. The interval **d4** is larger than an interval **d5** between the third connection region **38c** and the fourth connection region **38d**. Also, the intervals **d3**, **d4**, and **d5** are larger than an interval **d6** between the connection regions **38** in the central portion **62b**.

(57) Four trenches **22** are arranged between the first connection region **38a** and the second connection region **38b**, and three trenches **22** are arranged between the second connection region **38b** and the third connection region **38c**. Further, two trenches **22** are arranged between the third connection region **38c** and the fourth connection region **38d**. The interval between any two adjacent trenches **22** is substantially equal between the outer side portion **62a** and the central portion **62b**.

(58) When a forward biased voltage is applied to the body diode, the amount of holes accumulated in the drift region **34** decreases toward the end of the element region **62**. In the semiconductor device **100** of the second embodiment, the intervals between the connection regions **38** are gradually increased in the outer side portion **62a** from the central portion **62b** toward the peripheral region **64**. In the second embodiment, since the connection regions **38** are arranged more sparsely than in the first embodiment, the on-resistance can be further reduced. Moreover, since the intervals **d3** to **d5** of the connection regions **38** are adjusted according to the distribution of the amount of holes accumulated in the drift region **34**, the switching loss can be efficiently reduced.

Third Embodiment

(59) A semiconductor device **200** of a third embodiment differs from the semiconductor device **10** of the first embodiment in that multiple p-type auxiliary connection regions **39** are additionally provided in the outer side portion **62a**. As shown in FIGS. 6 and 7, the auxiliary connection regions **39** are arranged in a range defined between the two trenches **22** in the outer side portion **62a**. As

shown in FIG. 6, the auxiliary connection regions 39 are arranged at intervals in the y direction. As shown in FIG. 7, each auxiliary connection region 39 connects the body region 32 and the bottom region 36. As shown in FIG. 8, the auxiliary connection region 39 is not arranged in the section where the bottom region 36 is not arranged.

(60) In the third embodiment, when a reverse biased voltage is applied to the body diode, the holes flow from the bottom regions 36 to the body region 32 via the connection regions 38 as well as the auxiliary connection regions 39 in the outer side portion 62a. In the third embodiment, the holes can flow while branching in more paths than that in the first embodiment. Thus, the switching loss can be further reduced. The auxiliary connection regions 39 are arranged at intervals in the y direction. Specifically, as shown in FIGS. 6 and 7, the auxiliary connection regions 39 are arranged only in the areas directly above the bottom regions 36. Therefore, when the semiconductor device 200 is in the off state, the depletion layer extending from the auxiliary connection regions 39 to the drift region 34 does not extend widely in the drift region 34. As such, even if the auxiliary connection regions 39 exist, the path of the main current is less likely to be restricted when the semiconductor device 200 is turned on. Further, the auxiliary connection regions 39 are arranged only in the areas directly above the bottom regions 36 that do not function as the main current path when the semiconductor device 200 is turned on. Therefore, even if the auxiliary connection regions 39 exist, the on-resistance hardly increases. In the semiconductor device 200 of the third embodiment, as described above, the switching loss can be further reduced while suppressing the increase in on-resistance.

Fourth Embodiment

(61) A semiconductor device 300 of a fourth embodiment differs from the semiconductor device 10 of the first embodiment in that the interval of the trenches 22 in the outer side portion 62a is different. As shown in FIG. 9, in the fourth embodiment, three trenches 22 are arranged in a range defined between the two connection regions 38 in the outer side portion 62a. An interval D1 between the trenches 22 in the outer side portion 62a is smaller than an interval D2 between the trenches 22 in the central portion 62b. As shown in FIG. 10, the source region 30, the contact region 31, the body region 32 and the drift region 34 are provided in a semiconductor region defined between two adjacent trenches 22 in the outer side portion 62a.

(62) In the fourth embodiment, since the interval D1 between the trenches 22 in the outer side portion 62a is small, that is, the trenches 22 are densely arranged in the outer side portion 62a, the channel density of the outer side portion 62a is higher than that in the first embodiment. In the fourth embodiment, therefore, the channel resistance, that is, the on-resistance can be further reduced. Also in the present embodiment, since the interval between the connection regions 38 of the outer side portion 62a is larger than the interval between the connection regions 38 of the central portion 62b, the path of the main current is less likely to be restricted in the outer side portion 62a. Therefore, even if the interval D1 between the trenches 22 is reduced in the outer side portion 62a, the on-resistance due to the existence of the connection regions 38 hardly increases.

(63) Although the embodiments have been described in detail above, these are merely examples and do not limit the scope of claims. The techniques described in the claims include various modifications and modifications of the specific examples illustrated above. The embodiments described hereinabove will be modified as follows.

(64) (Modifications)

(65) In each of the embodiments described above, the connection region 38 is arranged in the area between the trenches 22. In other words, the connection region 38 is spaced apart from the trenches 22. As a modification, as shown in FIG. 11, the connection region 38 may be arranged at a position in contact with the side surface of the trench 22. The connection region 38 may be in contact with the gate insulating film 24, which is disposed in the trench 22, at a position below the body region 32. The connection region 38 may extend in the y direction along the side surface of the trench 22. Further, in this modification, the contact region 31 may be arranged at a position in contact with the

side surface of the trench 22. The contact region 31 may be in contact with the gate insulating film 24, which is disposed in the trench 22, on the side surface of the trench 22.

(66) In each of the embodiments described above, the bottom region 36 extends in the direction perpendicular to the longitudinal direction (y direction) of the trench 22. As another modification, as shown in FIG. 12, the bottom region 36 may extend parallel to the trench 22, that is, extends along the longitudinal direction of the trench 22. The bottom region 36 may be spaced from the bottom surface of the trench 22 and extend along the bottom surface of the trench 22. The bottom region 36 may be arranged to spaced apart in a direction (x direction) perpendicular to the longitudinal direction of the trenches 22.

(67) While only the exemplary embodiments and examples have been chosen to illustrate the present disclosure, it will be apparent to those skilled in the art from the specification and drawings that various changes and modifications can be made therein without departing from the scope of the disclosure as defined in the appended claims. Furthermore, the foregoing description of the exemplary embodiments and examples according to the present disclosure is provided for illustration only, and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents. The technical elements described in the present specification or the drawings exhibit technical usefulness alone or in various combinations, and are not limited to the combinations described in the claims at the time of filing. In addition, the techniques illustrated in the present specification or drawings achieve multiple objectives at the same time, and achieving one of the objectives itself has technical usefulness.

Claims

1. A semiconductor device comprising: a semiconductor substrate having an element region and a peripheral region disposed on a periphery of the element region; a plurality of trenches defined on an upper surface of the semiconductor substrate, each of the plurality of trenches extending in a first direction along the upper surface of the semiconductor substrate, the plurality of trenches being arranged spaced apart from each other in a second direction that is along the upper surface of the semiconductor substrate and perpendicular to the first direction; a gate insulating film covering an inner surface of each of the plurality of trenches; and a gate electrode disposed in each of the plurality of trenches and insulated from the semiconductor substrate by the gate insulating film, wherein the element region includes: an n-type source region that is exposed on the upper surface of the semiconductor substrate and is in contact with the gate insulating film disposed in a corresponding trench; a p-type contact region that is exposed on the upper surface of the semiconductor substrate; a p-type body region that is in contact with the gate insulating film disposed in the corresponding trench at a position below the source region, and is in contact with the contact region; an n-type drift region that is in contact with the gate insulating film disposed in the corresponding trench at a position below the body region, and is separated from the source region by the body region; a p-type bottom region that is disposed below the corresponding trench and spaced from a bottom surface of the corresponding trench, and is surrounded by the drift region; and a plurality of p-type connection regions that connects the body region and the bottom region, each of the plurality of p-type connection regions extending in the first direction, and the plurality of p-type connection regions being arranged spaced apart from each other in the second direction, wherein the element region has outer side portions at opposite ends of the element region in the second direction, and a central portion defined between the outer side portions, and wherein an interval in the second direction between the connection regions provided in at least one of the outer side portions is greater than an interval in the second direction between the connection regions provided in the central portion.

2. The semiconductor device according to claim 1, wherein the interval in the second direction between the connection regions provided in the at least one of the outer side portions increases

toward a corresponding end of the element region.

3. The semiconductor device according to claim 1, wherein a plurality of the bottom regions extends in the second direction, and is arranged spaced apart from each other in the first direction.
 4. The semiconductor device according to claim 3, wherein the at least one of the outer side portions further includes a plurality of p-type auxiliary connection regions arranged spaced apart from each other in the first direction, and wherein each of the plurality of p-type auxiliary connection regions connects the body region and a corresponding bottom region.
 5. The semiconductor device according to claim 1, wherein an interval in the second direction between the trenches provided in the at least one of the outer side portions is smaller than an interval in the second direction between the trenches provided in the central portion.
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